Jan 2022 Q4 Sample Solutions.

Q4 a) $A = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$ (a, b, c, d) $B = \begin{bmatrix} 9 & 12 \\ 4 & 5 \end{bmatrix}$

$AX = B$

$\therefore X = A^{-1}B$ got $A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

(inverse of a 2×2)

$$\det A = ad - bc$$
$$= (5)(1) - (2)(2)$$
$$= 5 - 4$$
$$= 1$$

$$A^{-1} = \frac{1}{1}\begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}\begin{bmatrix} 9 & 12 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} (1)(9)+(-2)(4) & (1)(12)+(-2)(5) \\ (-2)(9)+(5)(4) & (-2)(12)+(5)(5) \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

b) $A = \begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

/ Identity Matrix

ii) determinant of A (3×3 matrix)

$$\det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

a_{11}: $\begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = (0)(1) - (1)(1) = -1$

$M_{11} = -1$ $\quad i=1, j=1$

$\therefore C_{11} = M_{11} = -1$ $\quad 1+1 = (2)$ even

a_{12}: $\begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $\begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} = (3)(1) - (1)(1) = 2$

$M_{12} = 2$ $\quad i=1, j=2$

$\therefore C_{12} = -M_{12} = -2$ $\quad 1+2 = (3)$ odd

a_{13}: $\begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $\begin{vmatrix} 3 & 0 \\ 1 & 1 \end{vmatrix} = (3)(1) - (0)(1) = 3$

$M_{13} = 3$ $\quad i=1, j=3$

$\therefore C_{13} = M_{13} = 3$ $\quad 1+3 = (4)$ even

$a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

$\det A = (1)(-1) + (3)(-2) + (2)(3)$

$= -1 - 6 + 6$

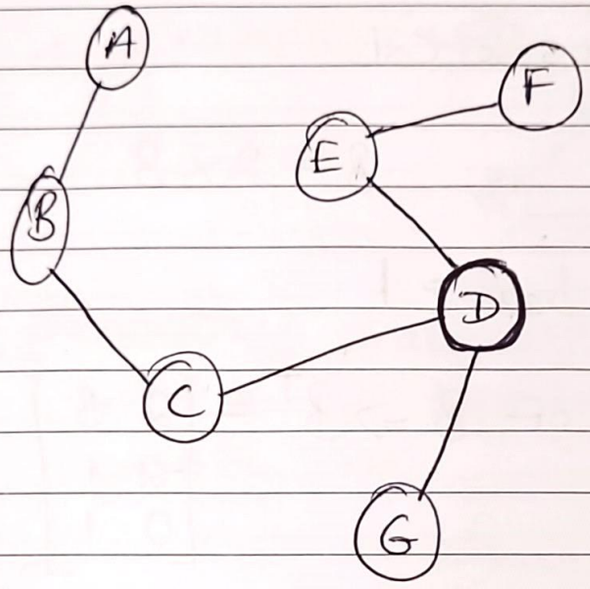
$= \boxed{-1}$

OR use

1	3	2	1	3	$(1\times0\times1) + (3\times1\times1) + (2\times3\times1)$
3	0	1	3	0	$= -$
1	1	1	1	1	$(1\times0\times2) + (1\times1\times1) + (1\times3\times3)$

$= 9 - 10 = \boxed{-1}$

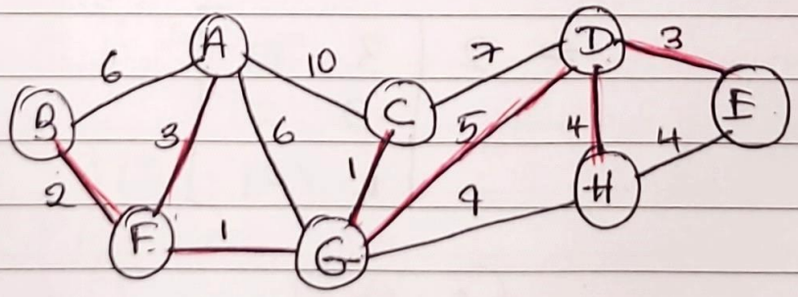
c) explain each method: depth first traversal & breadth first traversal (BFS)
(see Notes)



BFS: uses a Queue. ⟶ Queue: C ~~E~~ ~~G~~ ~~B~~ ~~F~~

Output: D C E G B F A
BFS

d) Prim's Algorithm:



T = {A, af, F, fg, G, gc, C, fb, B, gd, D, de, E, dh, H}
3 + 1 + 1 + 2 + 5 + 3 + 4

total cost = 19